

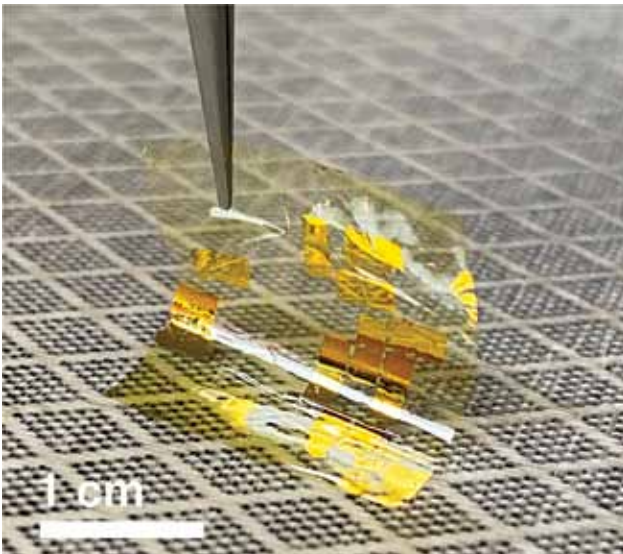


The VeCare platform comprises (clockwise from bottom left) a chip, wound sensor, bandage and app for real-time, point-of-care chronic wound monitoring (Credit: <https://news.nus.edu.sg>)

hence enabling fast and accurate wound assessment. Called VeCare, the point-of-care wound assessment platform consists of an innovative wound sensing bandage, an electronic chip, and a mobile app. The bandage comprises a wound contact layer, a breathable outer barrier, a micro-fluidic wound fluid collector, and a flexible immunosensor.

New material for ultrathin, lightweight solar panels

Researchers at Stanford University have a new group of photovoltaic materials that can be used in applications ranging from self-powered wearable devices and sensors to lightweight aircraft and electric vehicles. Chief among

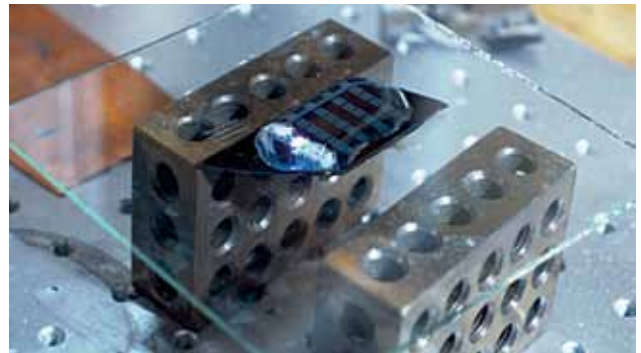


Transition metal dichalcogenide solar cells on a flexible polyimide substrate (Credit: <https://news.stanford.edu>)

the benefits of these transition metal dichalcogenides—or TMDs—is that they absorb ultrahigh levels of the sunlight that strikes their surface compared to other solar materials. The prototype achieves 5.1% power conversion efficiency and could practically reach 27% efficiency upon optical and electrical optimisations. The biggest benefit, however, is the remarkable thinness, which not only minimises the material usage and cost but also makes TMD solar cells lightweight and flexible, and capable of being moulded to irregular shapes.

Low-cost, self-fabrication of silicon wafers made possible

Zachary Tong on his YouTube channel Breaking Taps recently experimented doping a silicon wafer with phosphoric acid and a fibre laser that's near-infrared using a process called thermal diffusion to fabricate a silicon wafer at home. He used some advanced equipment—including



Doping liquid phosphoric acid on top of the silicon wafer and embedding it with precise laser pulses (Credit: Breaking Taps)

ing a laser, a powerful electron microscope, and high-end software—for scanning probe microscopy purposes, etc, to accomplish the feat. UV laser was preferred over infrared laser while doping as the former tends to have less optical penetration, thereby damaging less silicon wafer. The aim of the experiment was to inspire others to fabricate their own semiconductors at low cost.

Robotic hand picks up fruits and veggies delicately but firmly

Engineers at Stanford University have developed a robotic hand, called farmhand, capable of handling various items from raw eggs and grape bunches to jugs and basketballs. The robotic hand has gecko-like adhesive on the inner surface of its grippers that creates a strong grip via microscopic flaps, thus applying no extra inward force to hold an object. On coming in contact with an object's exterior surface, the flaps create Van der Waals forces, allowing the